INCH-POUND

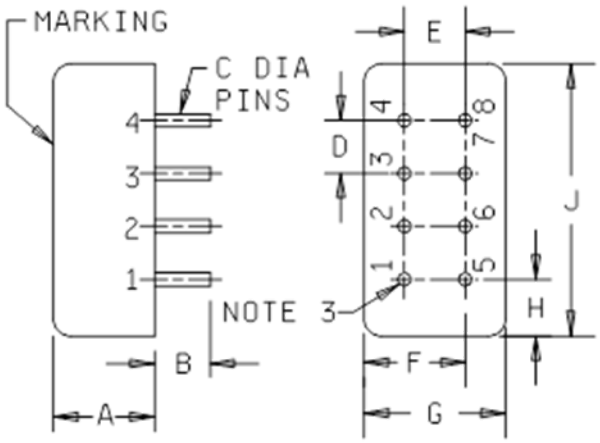
MIL-DTL-28837/1D
w/AMENDMENT 1
24 March 2022
SUPERSEDING
MIL-DTL-28837/1D
5 July 2012

DETAIL SPECIFICATION SHEET

MIXER STAGES, RADIO FREQUENCY, DOUBLE BALANCED, PIN PLUG-IN TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-28837.



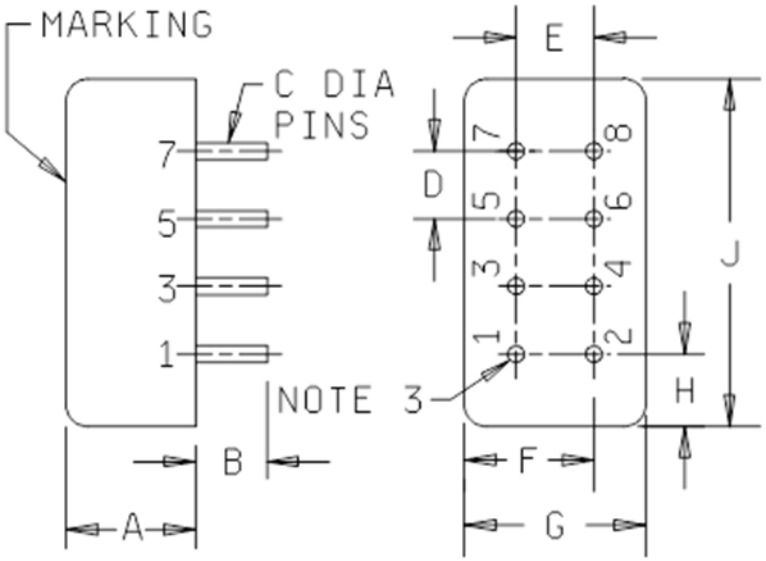
Dimension				
Ltr	Inches Min	Inches Max	mm Min	mm Max
A	.190	.210	4.83	5.33
B	.165	.205	4.19	5.21
C	.028 DIA	.032 DIA	0.71 DIA	0.81 DIA
D	.195	.205	4.95	5.21
E	.195	.205	4.95	5.21
F	.080	.110	2.03	2.79
G	.370	.400	9.40	10.16
H	.080	.110	2.03	2.79
J	.770	.800	19.60	20.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin number 1 shall have a contrasting color insulator, all pin numbers are to be marked on sides of case adjacent to the pins. The marking on the header is for reference only.

Termination for PIN M28837/01-01 and M28837/01-13	Termination for PIN M28837/01-02 and M28837/1-14
LO - Pins 1 and 5 RF - Pins 4 and 8 IF is formed by externally connecting pins 2 and 6 together and 3 and 7 together	LO - 1 and 5 RF - Pins 4 and 8 Pins 2, 4, and 6 are internally connected to case. IF is formed by externally connecting pins 3 and 7 together and using the case for reference.

FIGURE 1. Outline drawing for mixers PIN M28837/1-01, M28837/1-02, M28837/1-13, and M28837/1-14.

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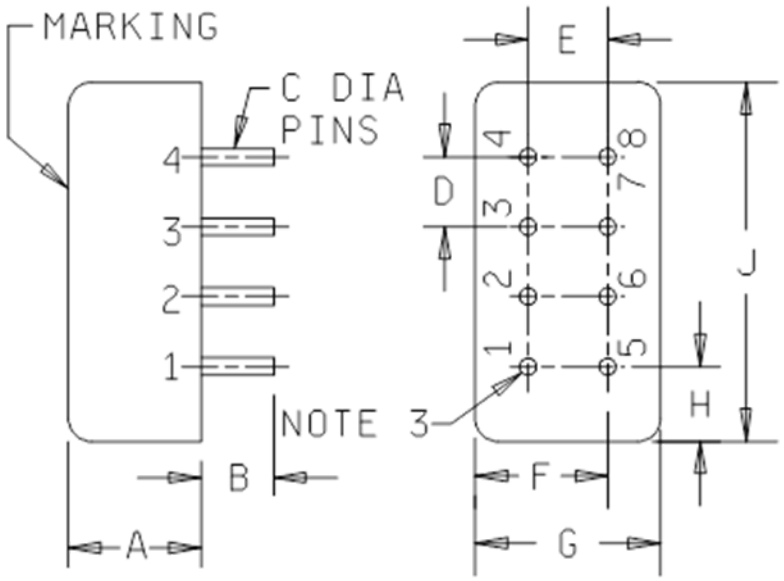
Dimension				
Ltr	Inches	Inches	mm	mm
	Min	Max	Min	Max
A	---	.400	---	10.16
B	.165	.205	4.19	5.21
C	.028 DIA	.032 DIA	0.71 DIA	0.81 DIA
D	.195	.205	4.95	5.21
E	.195	.205	4.95	5.21
F	.080	.110	2.03	2.79
G	.370	.400	9.40	10.16
H	.080	.110	2.03	2.79
J	.770	.800	19.60	20.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin number 1 shall have a contrasting color insulator, all pin numbers are to be marked on sides of case adjacent to the pins. The marking on the header is for reference only.

Termination for PIN M28837/01-03, -
M28837/01-08, and M28837/01-09
LO - Pins 7 and 8
RF - Pins 1 and 2
IF is formed by externally
connecting pins 3 and 4 together
for the grounded side.

FIGURE 2. Outline drawing for mixers PIN M28837/1-03, M28837/1-08, and M28837/1-09.

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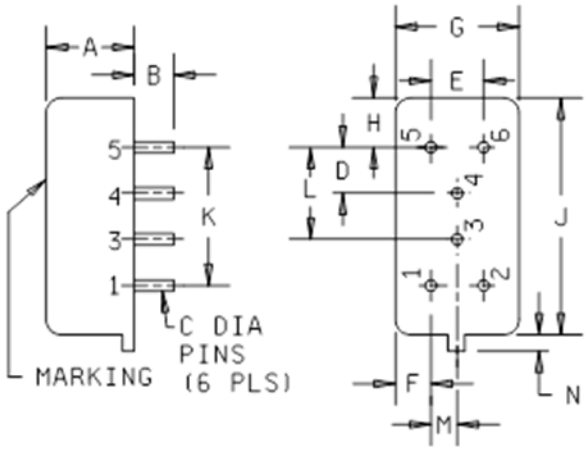
Dimension				
Ltr	Inches	Inches	mm	mm
	Min	Max	Min	Max
A	.400	.425	10.16	10.80
B	.165	.205	4.19	5.21
C	.028 DIA	.032 DIA	0.71 DIA	0.81 DIA
D	.195	.205	4.95	5.21
E	.195	.205	4.95	5.21
F	.080	.110	2.03	2.79
G	.370	.400	9.40	10.16
H	.080	.110	2.03	2.79
J	.770	.800	19.60	20.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin number 1 shall have a contrasting color insulator, all pin numbers are to be marked on sides of case adjacent to the pins. The marking on the header is for reference only.

Termination for PIN M28837/01-04 and M28837/01-12 LO - Pins 1 and 5 RF - Pins 4 and 8 IF is formed by externally connecting pins 3 and 7 together for the grounded side and using pins 2 and 6 for the ground side.

FIGURE 3. Outline drawing for mixers PIN M28837/1-04 and M28837/1-12.

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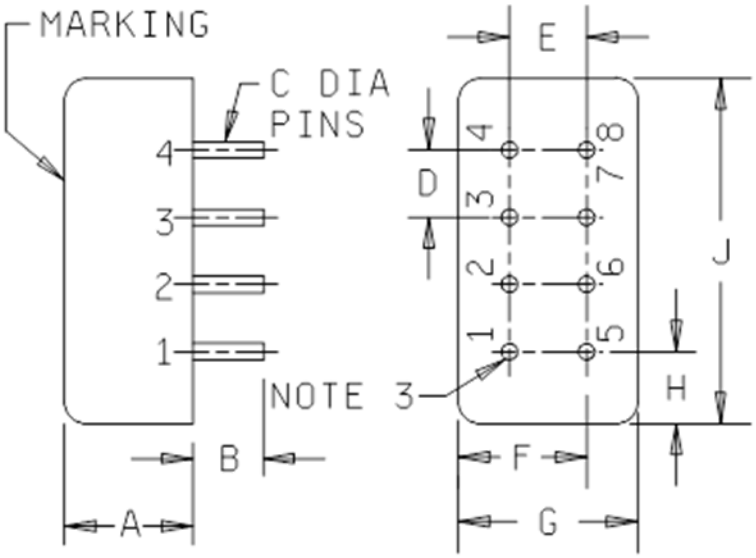
Dimension				
Ltr	Inches	Inches	mm	mm
	Min	Max	Min	Max
A	.390	.430	9.91	10.92
B	.100	.130	2.54	3.30
C	.023 DIA	.027 DIA	0.58 DIA	0.69 DIA
D	.295	.305	7.49	7.75
E	.295	.305	7.49	7.75
F	.090	.130	2.29	3.30
G	.500	.525	12.7	13.34
H	.090	.130	2.29	3.30
J	1.000	1.025	25.4	26.04
K	.795	.805	20.19	20.45
L	.495	.505	12.57	12.83
M	.145	.155	3.68	3.94
N	---	.040	---	1.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin number 1 shall have a contrasting color insulator, all pin numbers are to be marked on sides of case adjacent to the pins. The marking on the header is for reference only.

Termination for PIN 28837/1-05, M28837/1-06, M28837/1-07 and M28837/1-10
LO - Pins 1 and 2
RF - Pins 5 and 6
IF - Pins 3 and 4.

FIGURE 4. Outline drawing for mixers PIN M28837/1-05, M28837/1-06, M28837/1-07 and M28837/1-10.

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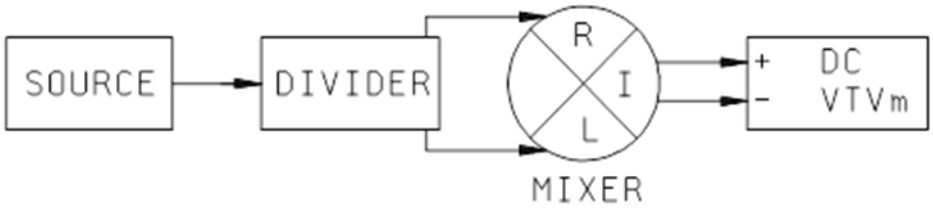
Dimension				
Ltr	Inches	Inches	mm	mm
	Min	Max	Min	Max
A	.297	.327	7.54	8.31
B	.188	.218	4.78	5.54
C	.028 DIA	.032 DIA	0.71 DIA	0.81 DIA
D	.195	.205	4.95	5.21
E	.195	.205	4.95	5.21
F	.080	.110	2.03	2.79
G	.370	.400	9.40	10.16
H	.080	.110	2.03	2.79
J	.775	.805	19.69	20.45

Termination for PIN M28837/1-11 LO - Pins 4 and 8 RF - Pins 1 and 5 IF - Pins 3 and 7 Pins 2, 4, and 6 are grounded	Termination for PIN M28837/1-15 and M28837/1-16 LO - Pin 5 RF - Pin 8 IF - Pins 3 and 7 All other pins are grounded.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin number 1 shall have a contrasting color insulator, all pin numbers are to be marked on sides of case adjacent to the pins. The marking on the header is for reference only.

FIGURE 5. Outline drawing for mixers PIN M28837/1-11, M28837/1-15, and M28837/1-16.

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NOTE: The phase balance test shall be performed in accordance with figure 6. When any difference from the mean reading (approximately -0.35 volts) on the VTVM is greater than 0.05 volts, the mixer fails this test.

FIGURE 6. Phase balance test setup for PIN M28837/1-01 and M28837/1-02.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1 through 6.

Electrical characteristics:

Operating frequency range: See table I.

LO drive power: See table I.

Conversion loss (max): See table I.

Noise figure (SSB): See table I.

Isolation (minimum): See table I.

Maximum input power (rms): See table I.

Conversion compression (max): See table I.

Desensitization (max): See table I.

DC relative polarity: See table I.

VSWR: See table I.

Third order two-tone intermodulation: See table I.

Impedance: 50 ohms.

Physical and environmental characteristics:

Weight: See table II.

Temperature range: See table II.

Mechanical shock: See table II.

Hermetic seal: See table II.

Terminal strength: See table II.

Life: See table II.

Part or Identifying Number (PIN)
Example of Nonscreened PIN M28837/1- 01N
Example of Screened PIN M28837/1- 01S
Space flight mixers shall be marked with “T”, M28837/1- 01 T

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TABLE I. Physical and environmental characteristics. 1/

Dash no.	Operating frequency range (MHz)	LO input drive power in dBm 2/	Maximum conversion loss (SSB) dB MHz	Noise figure (SSB) dB MHz	Isolation			
					Minimum			Frequency MHz
					LO-RF dB	LO-IF dB	RF-IF dB	
01N 01S	LO .05-200 RF .05-200 IF DC-200	minimum +4 TV +7 maximum +13	6.5 at f_L and f_R 0.2-50 f_I DC-50	6.5 at f_L and f_R 1-50 f_I 0.4-50	45	40	N/A	.05-30
			8.0 at f_L and f_R 50-200 f_I DC-200	8.0 at f_L and f_R 50-200 f_I 0.4-200	35	30	N/A	30-200
			8.5 at f_L and f_R 0.05-0.2 f_I DC-0.4	N/A				
02N 02S	LO 5-500 RF 5-500 IF DC-500	minimum +4 TV +7 maximum +13	7.0 at f_L and f_R 5-150 f_I 0.4-150	7.0 at f_L and f_R 10-100 f_I 0.4-100	45 30	40 25	N/A N/A	5-50 50-500
			9.0 at f_L and f_R 150-500 f_I 0.4-500	9.0 at f_L and f_R 100-500 f_I 0.4-500				
03N 03S	LO .5-500 RF .5-500 IF DC-500	TV +7	7.5 at f_L and f_R 1-250 f_I DC-250	N/A	35 30 25	30 25 20	N/A N/A N/A	0.5-1 1-250 250-500
			8.5 at f_L and f_R 0.5-500 f_I DC-500					

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued. 1/

Dash no.	RF and LO 4/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 5/	VSWR (maximum)			Third order, two tone intermodulation	
					LO	IF	RF	dBm	Frequencies
01N 01S	50 mW	N/A	N/A	Negative					
02N 02S	400 mW	N/A	N/A	Negative					
03N 03S	350 mW	1.0 dB at P_R level of 0 dBm	1.0 dB at P_{R2} level of -2 dBm	Negative					

See footnotes at end of table.

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TABLE I. Physical and environmental characteristics - Continued. 1/

Dash no.	Operating frequency range (MHz)	LO input drive power in dBm 2/	Maximum conversion loss (SSB) dB MHz	Noise figure (SSB) dB MHz	Isolation Minimum LO-RF dB	Isolation Minimum LO-IF dB	Isolation Minimum RF-IF dB	Isolation Frequency MHz
04N 04S	LO 2-500 RF 2-400 IF DC-800	minimum +10 TV +20 maximum +23	f_L 5-200 f_R 20-80 6.5 at f_I 5-80 7.0 at f_I .4-120	3/ f_L 5-200 f_R 20-80 6.5 at f_I 5-80 7.0 at f_I DC-120	40	40	N/A	2-32
			f_L 5-300 f_R 5-300 7.0 at f_I 5-80 8.0 at f_I DC-300	3/ f_L 5-300 f_R 5-300 7.0 at f_I 5-80 8.0 at f_I 2-120	35	35	N/A	32-100
			f_L 5-470 f_R 5-400 7.5 at f_I 5-80	3/ f_L 5-470 f_R 5-400 8.0 at f_I 5-80	25	25	N/A	100-500
			f_L 2-500 f_R 2-400 9.0 at f_I DC-800	3/ f_L 2-500 f_R 2-400 9.5 at f_I 5-800	N/A	N/A	17	2-400
05N	LO .05-200	minimum +7	7.5 at f_L and f_R 0.2-50 f_I DC-50	7.5 at f_L and f_R 1-50 f_I 0.4-50	45 30	40 25	N/A N/A	.05-30 30-200
05S	RF .05-200 IF DC-200	TV +13 maximum +17	9.0 at f_L and f_R .05-200 f_I DC-200	9.0 at f_L and f_R 50-200 f_I 0.4-200	N/A	N/A	24	.05-3
06N 06S	LO .5-500 RF .5-500 IF DC-500	minimum +7 TV +17 maximum +20	7.0 at f_L and f_R 0.5-30 f_I DC-30 7.5 at f_L and f_R 30-100 f_I DC-100 9.0 at f_L and f_R 100-500 f_I DC-500	7.0 at f_L and f_R 1-30 f_I 0.4-30 7.5 at f_L and f_R 30-100 f_I 0.4-100 9.0 at f_L and f_R 100-500 f_I 0.4-500	55 45 35 N/A	45 35 25 N/A	N/A N/A N/A 20	0.5-30 30-100 100-500 0.5-500

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued. 1/

Dash no.	RF and LO 4/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 5/	VSWR (maximum)			Third order, two tone intermodulation	
					LO	IF	RF	dBm	Frequencies
04N 04S	200 mW	1.0 dB at P_R +15 dBm P_L +23 dBm	1.0 dB at P_{R2} +13 dBm P_L +23 dBm	Positive					
05N 05S	70 mW	1.0 dB at P_R +10 dBm P_L +17 dBm	1.0 dB at P_{R2} +7 dBm P_L +17 dBm	Negative					
06N 06S	140 mW	1.0 dB at P_R +8 dBm P_L +17 dBm	1.0 dB at P_{R2} +7 dBm P_L +17 dBm	Negative					

See footnotes at end of table.

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TABLE I. Physical and environmental characteristics - Continued. 1/

Dash no.	Operating frequency range (MHz)	LO input drive power in dBm 2/	Maximum conversion loss (SSB) dB MHz	Noise figure (SSB) dB MHz	Isolation			
					Minimum			Frequency MHz
					LO-RF dB	LO-IF dB	RF-IF dB	
07N 07S	LO 5-400 RF 5-400 IF DC-400	TV +17	7.5 at f_L and f_R 5-200 f_I DC-200 9.0 at f_L and f_R 200-400 f_I DC-400	N/A	35 25	30 20	N/A N/A	5-100 100-400
08N 08S	LO 0.5-500 RF 0.5-500 IF DC-500	TV +17	7.5 at f_L and f_R 0.5-5 8.5 at f_L and f_R 5-500 f_I DC-500	N/A	45 30 25	35 30 20	N/A N/A N/A	0.5-5 5-250 250-500
09N 09S	LO 1-750 RF 1-750 IF DC-750	TV +7	7.5 at f_L and f_R 2-375 f_I DC-375 8.5 at f_L and f_R 1-750 f_I DC-750	N/A	45 30 25	30 25 20	N/A N/A N/A	1-2 2-375 375-750
10N 10S	LO .4-500 RF .4-500 IF DC-500	minimum +7 TV +13 maximum +17	7.5 at f_L and f_R 1-50 f_I 0.4-50 9.0 at f_L and f_R 0.4-500 f_I 0.4-500	7.5 at f_L and f_R 1-50 f_I 0.4-50 9.0 at f_L and f_R 50-500 f_I 0.4-500	45 25	40 25	N/A N/A	0.4-50 50-500

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued. 1/

Dash no.	RF and LO 4/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 5/	VSWR (maximum)			Third order, two tone intermodulation	
					LO	IF	RF	dBm	Frequencies
07N 07S	35 mW	N/A	N/A	Negative					
08N 08S	350 mW	1.0 dB at P_R +10 dBm	N/A	Negative					
09N 09S	350 mW	1.0 dB at P_R +0 dBm	1.0 dB at P_{R2} -2 dBm	Negative					
10N 10S	70 mW	1.0 dB at P_R +10 dBm P_L +17 dBm	1.0 dB at P_{R2} +7 dBm P_L +17 dBm	Negative					

See footnotes at end of table.

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TABLE I. Physical and environmental characteristics - Continued. 1/

Dash no.	Operating frequency range (MHz)	LO input drive power in dBm 2/	Maximum conversion loss (SSB) dB MHz	Noise figure (SSB) dB MHz	Isolation			
					Minimum			Frequency MHz
					LO-RF	LO-IF	RF-IF	
					dB	dB	dB	
11N 11S	LO 1-500 RF 1-500 IF DC-500	TV +23	8.5 at f_L and f_R 2-400 f_I DC-400	Within 1 dB of conversion loss	50 40 30 20	40 30 20 20	25 20 20 20	1-100 100-200 200-300 300-500
			9.5 at f_L and f_R 1-500 f_I DC-500					
12N 12S	LO .002-12 RF .002-12 IF DC-12	minimum +4 TV +7 maximum +13	6.0 at f_L and f_R .01-5 f_I 0.4-5	6.0 at f_L and f_R .01-5 f_I 0.4-5	45	40	N/A	.002-5 5-12
			8.0 at f_L and f_R .002-12 f_I 0.4-12	8.0 at f_L and f_R 5-12 f_I 0.4-12	40	30	N/A	
13N 13S	LO .05-200 RF .05-200 IF DC-200	minimum -2 TV 0.0 maximum +13	6.5 at f_L and f_R 0.2-50 f_I DC-50	6.5 at f_L and f_R 0.2-50 f_I 0.4-50	45	40	N/A	.05-30
			8.0 at f_L and f_R 50-200 f_I DC-200	8.0 at f_L and f_R 50-200 f_I 0.4-200	35	30	N/A	30-200
			8.5 at f_L and f_R 0.05-0.2 f_I DC-0.2					

See footnotes at end of table.

TABLE I. Electrical characteristics - Continued. 1/

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Dash no.	RF and LO 4/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 5/	VSWR (maximum)			Third order,	two tone intermodulation
					LO	IF	RF	dBm	Frequencies
11N 11S	425 mW	1.0 dB at P_R +14 dBm P_L +23 dBm	1.0 dB at P_{R2} +12 dBm P_L +23 dBm	Negative					
12N 12S	35 mW	N/A	N/A	Negative					
13N 13S	50 mW	N/A	N/A	Negative	2.5:1	2.0:1	2.5:1	6/ +5 +3	f_{L0}: 200 MHz f_{R1}: 150 MHz f_{R2}: 155 MHz at -10 dBm -15 dBm

See footnotes at end of table.

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Check the source to verify that this is the current version before use.

TABLE I. Physical and environmental characteristics - Continued. 1/

Dash no.	Operating frequency range (MHz)	LO input drive power in dBm 2/	Maximum conversion loss (SSB) dB MHz	Noise figure (SSB) dB MHz	Isolation			
					Minimum			Frequency MHz
					LO-RF dB	LO-IF dB	RF-IF dB	
14N 14S	LO 5-500 RF 5-500 IF DC-500	minimum -2 TV 0.0 maximum +13	7.0 at f_L and f_R 10-100 f_I 10-100	7.0 at f_L and f_R 10-100 f_I 10-100	45	40	N/A	5-50
			8.0 at f_L and f_R 100-200 f_I 10-200	8.0 at f_L and f_R 100-200 f_I 10-200	30	25	N/A	50-500
			9.5 at f_L and f_R 5-500 f_I .5-500	9.5 at f_L and f_R 5-500 f_I .5-500				
15N 15S	LO 1-500 RF 1-500 IF DC-500	minimum +3 TV +7 maximum +10	7.0 at f_L and f_R 1-400 f_I DC-400	Within 1 dB of conversion loss	40	35	N/A	1-100
			8.0 at f_L and f_R 1-500 f_I DC-500		30	25	N/A	100-500
16N 16S	LO 1-500 RF 1-500 IF DC-500	minimum +13 TV +15 maximum +20	7.0 at f_L and f_R 1-400 f_I DC-400	Within 1 dB of conversion loss	40	40	N/A	1-100
			8.0 at f_L and f_R 400-500 f_I DC-500		30	25	N/A	100-500

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued. 1/

Dash no.	RF and LO 4/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 5/	VSWR (maximum)			Third order, two tone intermodulation	
					LO	IF	RF	dBm	Frequencies
14N 14S	400 mW	N/A	N/A	Negative	2.5:1	2.0:1	2.5:1	6/ +4 +2	f_{L0}: 500 MHz f_{R1}: 400 MHz f_{R2}: 405 MHz at -10 dBm -15 dBm
15N 15S	300 mW	1.0 dB at P_R +3 dBm	1.0 dB at P_R +1 dBm	Positive	2.0:1	1.5:1	1.5:1	+12 7/	f_{L0}: 500 MHz f_{R1}: 400 MHz at -10 dBm f_{R2}: 401 MHz at -10 dBm
16N 16S	300 mW	1.0 dB at P_R +8 dBm	1.0 dB at P_R +1 dBm	Positive	2.0:1	2.0:1	2.0:1	+18 7/	f_{L0}: 500 MHz f_{R1}: 400 MHz at -10 dBm f_{R2}: 401 MHz at -10 dBm

1/ Where data does not appear in the table, the requirement does not apply.
2/ Unless otherwise specified, TV is the test value of the LO power for the electrical characteristics.
3/ Noise figure values apply over a temperature range of -54°C to 85°C. For a temperature range of 85°C to 100°C, the noise figure over the specified frequency band will be allowed an additional 0.5 dB tolerance.
4/ These values are for +25°C and are derated linearly to +125°C.
5/ With two-in phase signals applied to the LO and RF ports and unused pins grounded.
6/ Input intercept point.
7/ Output intercept point.

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TABLE II. Physical and environmental characteristics. 1/

Dash number	Weight	Temperature range in Celsius (operating on top nonoperating on bottom)	Mechanical shock (MIL-STD-202-213)	Hermetic seal (MIL-STD-202-112)	Terminal strength (MIL-STD-202-211)	Life
01N 01S	0.18 oz. (5.0 g)	-54° to +100° -65° to +100°	---	Yes	---	---
02N 02S	0.18 oz. (5.0 g)	-54° to +100° -65° to +100°	---	Yes	---	---
03N 03S	0.18 oz. (5.0 g)	-55° to +100° -55° to +100°	---	Yes	---	---
04N 04S	0.21 oz. (6.0 g)	-54° to +100° -65° to +100°	C	Yes	---	---
05N 05S	0.24 oz. (6.7 g)	-54° to +100° -65° to +100°	---	Yes	---	Storage 2 years minimum
06N 06S	0.25 oz. (7.2 g)	-54° to +100° -65° to +100°	C	Yes	---	100,000 hours
07N 07S	0.23 oz. (6.5 g)	-54° to +100° -65° to +100°	C	Yes	C	---
08N 08S	0.18 oz. (5.0 g)	-55° to +100° -55° to +100°	---	Yes	C	10 years with 300,000 hours continuous
09N 09S	0.18 oz. (5.0 g)	-55° to +100° -55° to +100°	---	Yes	---	---
10N 10S	0.24 oz. (6.7 g)	-54° to +100° -65° to +100°	C	Yes	C	50,000 hours
11N 11S	0.25 oz. (7.0 g)	-55° to +85° -55° to +85°	---	No	---	---
12N 12S	0.18 oz. (5.0 g)	-54° to +100° -65° to +100°	---	Yes	C	50,000 hours
13N 13S	0.18 oz. (5.0 g)	-54° to +100° -65° to +100°	C	Yes	C	---
14N 14S	0.18 oz. (5.0 g)	-54° to +100° -65° to +100°	C	Yes	C	---
15N 15S	0.25 oz. (7.0 g)	-54° to +100° -65° to +100°	C	Yes	---	50,000 hours
16N 16S	0.25 oz. (7.0 g)	-54° to +100° -65° to +100°	C	Yes	---	50,000 hours

1/ Where --- is indicated, reference requirements as set forth in MIL-DTL-28837.

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Referenced documents: In addition to MIL-DTL-28837, this document references the following
MIL-STD-202-112
MIL-STD-202-211
MIL-STD-202-213

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5895-2022-001)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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